

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-004: Student Input Field Validation for Add and Edit Operations

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Field Validation
Student Name	N. Bavanika
Student ID	JFICT24_19
Implementation Status	Completed
Date	8 October 2026

1. Change Request Summary

1.1 Business Requirement

The change was required to validate student information entered during Add and Edit operations so that incomplete, incorrectly formatted, or invalid student data is prevented from being submitted and stored. Validation is required at both frontend and backend levels.

1.2 Existing Behaviour

Before implementation:

- The administrator could add and edit student records through the existing student form.
- Only basic required-field checks were available before submission.
- Field-specific format validation was limited.
- Invalid registration numbers, names, phone numbers, email formats, and weak passwords could reach the backend if client-side checks were bypassed.
- Existing Add, Edit, Delete, authentication, authorization, and password hashing functionality was already available.

1.3 Requested Change

Student input validation was required for Add and Edit operations.

- Validate all required student fields before submission.
- Registration number must follow the REG001 format.
- First name and last name must contain at least 3 alphabetic characters.
- Email address must use a valid email format.
- Phone number must contain exactly 10 numeric digits.
- Password is required when creating a new student.
- Password must contain at least 8 characters, one uppercase letter, one lowercase letter, one number, and one special character.
- Display clear validation errors to the administrator.
- Prevent invalid frontend submissions.
- Apply the same validation on the server side to prevent client-side validation bypass.

- Preserve existing Add, Edit, Delete, authentication, authorization, and bcrypt password hashing behaviour.

2. Implementation

2.1 Implementation Summary

The CR-004 requirements were implemented in the existing student controller and student form modal. Frontend validation was added to provide immediate field-level feedback and prevent invalid submissions. Backend validation was added to independently validate incoming requests before database operations.

No database schema changes, new API endpoints, route changes, or authentication/authorization changes were required.

2.2 Affected Files

File	Change	Purpose
studentController.js	Modified	Added server-side validation for student Add and Edit operations, including registration number, names, email, phone number, and password rules.
StudentFormModal.jsx	Modified	Added client-side validation, field-specific validation errors, phone-number input restrictions, and validation before API submission.

Additional files: None

2.3 Technical Changes

Frontend Validation

- Added fieldErrors state to store individual validation messages.
- Added validation for registration number using the REG001 format.
- Added minimum 3-character and alphabetic-only validation for first and last names.
- Added email format validation.
- Added exactly 10-digit numeric validation for phone number.
- Restricted phone number input to numeric characters and a maximum of 10 digits.
- Added password validation for new students: minimum 8 characters, uppercase, lowercase, number, and special character.
- Validation runs before the API request is submitted.

Backend Validation

- Added a reusable validation function inside studentController.js.
- Applied validation to the Add Student operation.
- Applied validation to the Edit Student operation.
- Backend returns HTTP 400 with validation errors when invalid input is received.

- Validation is performed before database operations.
- Existing duplicate registration number and duplicate email checks remain unchanged.
- Existing bcrypt password hashing remains unchanged.

Existing Functionality Preservation

- Existing student model was not modified.
- Existing student routes were not modified.
- No database schema changes were made.
- Existing administrator authentication and authorization were preserved.
- Existing Add, Edit, and Delete operations were preserved.

2.4 Implementation Logic

1. Administrator opens the Add Student or Edit Student form.
2. Student input values are entered into the existing form.
3. Frontend validation checks each applicable field.
4. If validation fails, field-specific errors are displayed and the API request is not submitted.
5. If frontend validation passes, the existing API request is sent.
6. Backend validation independently checks the received data.
7. If backend validation fails, the request returns HTTP 400 with validation errors.
8. If validation passes, the existing database and password-handling logic continues.
9. Edit operation does not expose or update a password, so password validation applies only when creating a new student.

2.5 Error and Data Handling

Invalid input is rejected before database modification. The frontend displays field-specific validation messages for common input errors. The backend provides validation errors in the API response to prevent invalid data from being stored when client-side validation is bypassed.

3. Testing & Verification

3.1 Testing Approach

The implementation is prepared for manual frontend and backend verification.

Testing Types:

- Functional testing
- Positive testing
- Validation testing
- UI testing
- Backend/API validation testing
- Regression testing
- CRUD verification
- Authentication / authorization regression testing

Environment: Development

Browser: [Chrome / Edge]

Test Role: Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Submit Add Student with all fields empty	Required-field validation errors are displayed and request is blocked.	Required-field validation errors were displayed and submission was blocked.	Pass
TC-02	Enter invalid registration number such as REG01	Registration number format error is displayed and invalid request is rejected.	Registration number format error was displayed and invalid submission was rejected.	Pass
TC-03	Enter first/last name below 3 characters	Minimum length validation error is displayed.	Minimum length validation error was displayed and submission was blocked.	Pass
TC-04	Enter numbers/special characters in first or last name	Alphabetic-only validation error is displayed.	Alphabetic-only validation error was displayed and invalid submission was blocked.	Pass
TC-05	Enter invalid email format	Valid email format error is displayed.	Email validation error was displayed and invalid submission was blocked.	Pass
TC-06	Enter phone number with fewer/more than 10 digits	Phone number validation error is displayed and invalid input is rejected.	Phone number validation was applied and invalid length was rejected.	Pass
TC-07	Enter non-numeric characters in phone number	Non-numeric phone input is prevented/rejected.	Non-numeric characters were prevented in the phone number field.	Pass
TC-08	Enter weak password without required character types	Password validation error is displayed.	Password validation error was displayed for passwords that did not meet the required rules.	Pass
TC-09	Submit valid Add Student data	Student is created successfully and existing CRUD behaviour remains functional.	Student was created successfully and existing CRUD functionality continued to work.	Pass
TC-10	Edit student with invalid field data	Validation errors are displayed and update is blocked.	Validation errors were displayed and the	Pass

			invalid update was blocked.	
TC-11	Edit student with valid data	Student is updated successfully.	Student was updated successfully with valid data.	Pass
TC-12	Verify existing Delete operation after validation changes	Delete functionality continues to work correctly.	Student deletion continued to work correctly after the validation changes.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12
Failed	0
Pass Rate	100%

Overall Test Result: Pending Verification

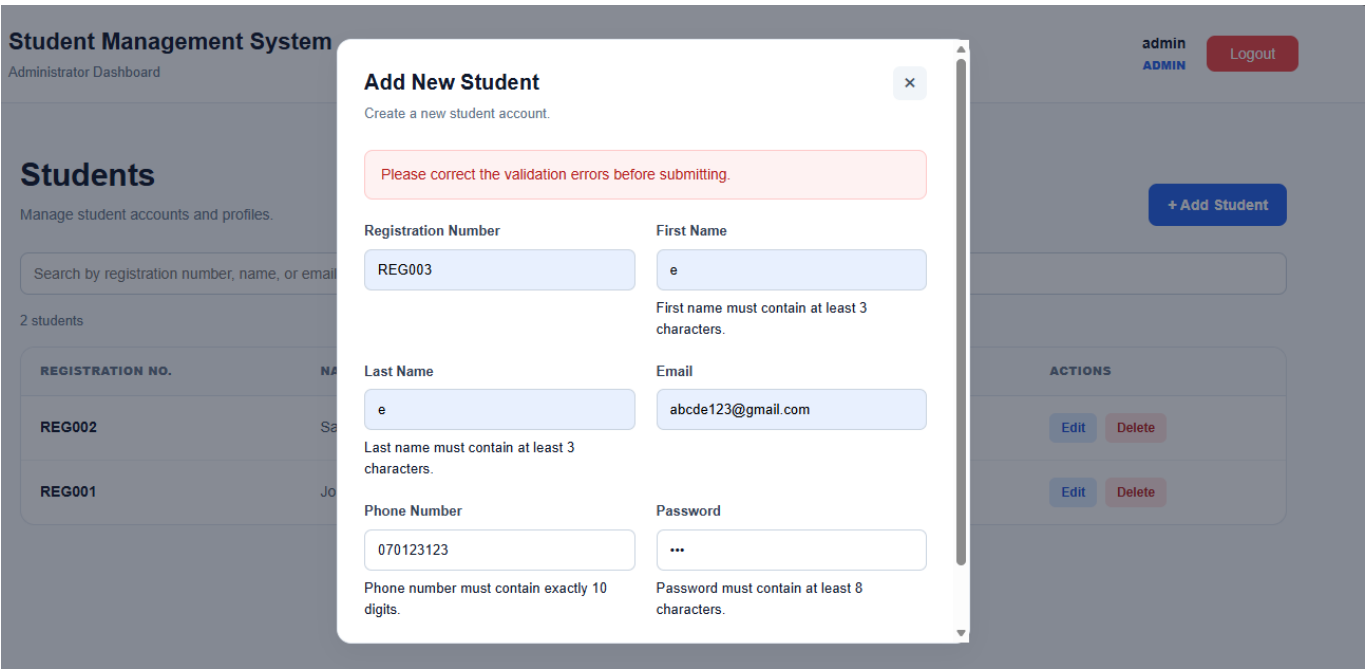
4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	All required student fields are validated before submission.	Frontend and backend validation testing	Met
AC-002	Registration number follows the REG001 format.	Registration number validation test	Met
AC-003	First and last names require at least 3 alphabetic characters.	Name validation test	Met
AC-004	Email address must use a valid format.	Email validation test	Met
AC-005	Phone number must contain exactly 10 numeric digits.	Phone validation test	Met
AC-006	New student password meets all required password rules.	Password validation test	Met
AC-007	Invalid frontend submissions are prevented and clear errors are displayed.	UI validation test	Met
AC-008	Backend validation prevents invalid requests from bypassing frontend validation.	Direct API validation test	Met
AC-009	Existing Add, Edit, Delete, authentication, authorization, and password hashing remain functional.	Regression testing	Met

5. Evidence

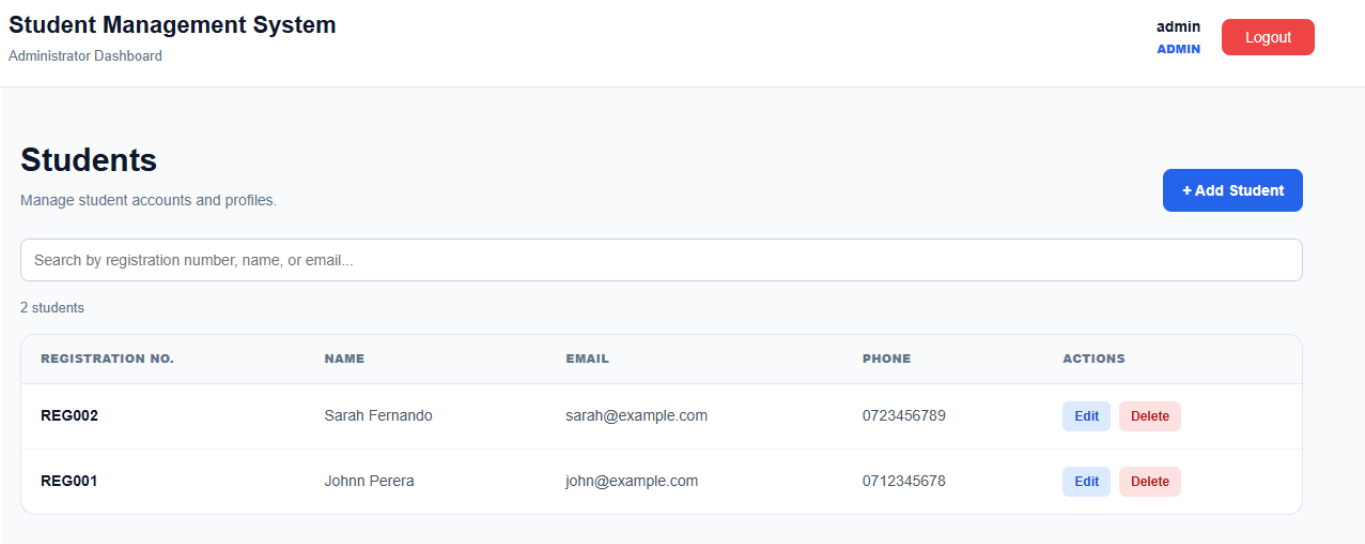
Evidence 01 – Student Add Form Validation

Related: AC-001, AC-002, AC-003, AC-004, AC-005, AC-006, AC-007



Evidence 02 – Student Edit

Related: AC-008, AC-009



6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing CRUD	Verified
Final Status	Accepted

6.1 Final Summary

CR-004 was implemented to provide student input field validation for Add and Edit operations. Frontend validation was added to StudentFormModal.jsx for registration number, names, email, phone number, and new-student password requirements. Backend validation was added to studentController.js to independently validate incoming requests and prevent client-side validation bypass. The implementation does not modify the database schema, student model, routes, authentication, authorization, or existing password hashing behaviour. Existing CRUD functionality is intended to remain unchanged. Formal testing and evidence verification are pending.

6.2 Issues / Limitations

No significant implementation issues identified. Formal test execution and screenshot evidence remain to be completed.

Student Name	N. Bavanika
Student ID	JFICT24_19
Date	8 October 2026
Signature	

~ END OF REPORT ~